

Analog Assignment - 2

Design of a 2-stage OTA

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Design Assumptions and Constraints

Technology: TSMC $0.18\mu\text{m}$ ($V_{dd} = 1.8\text{V}$, $L_{min} = 0.18\mu\text{m}$, $k'_{min} = 0.27\mu\text{m}$)

Target DC gain: $> 40\text{dB}$

V_{ov} (Gate Overdrive Voltage): 0.2V for all transistors.

Load Capacitance (C_L): Assumed ~~10~~ 10pF

Compensation Capacitance (C_c): Taken as 5pF

Process Parameters: $\mu_n C_{ox} = 230\mu\text{A}/\text{V}^2$ $V_{tn} = 0.37\text{V}$
 $\mu_p C_{ox} = 100\mu\text{A}/\text{V}^2$ $V_{tp} = 0.39\text{V}$

Current Budgeting

Tail Current: $100\mu\text{A}$

$$I_{D1} = I_{D2} = 50\mu\text{A}$$

$\Rightarrow$ Reference Current: $\frac{1}{10} \times 100 = 10\mu\text{A}$

Second Stage Current: $10 \times 50 = 500\mu\text{A}$

Since the dominant pole should be 5 times closer than the non-dominant pole for phase margin.

Initial Calculations

Considering $I_{tail} = 100\mu A$

$$I_{ref} = 10\mu A$$

$$I_{D1} = I_{D2} = 50\mu A$$

$$V_{ov} = 0.2V$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right) (V_{ov})^2$$

$$\Rightarrow \frac{W}{L} = \frac{2I_D}{\mu_n C_{ox} (V_{ov})^2} = \frac{2 \times 50\mu A}{230\mu A/V^2 \times (0.2)^2} = \underline{\underline{10.869}}$$

Similarly,

$$\frac{W}{L} = \frac{2I_D}{\mu_p C_{ox} (V_{ov})^2} = \frac{2 \times 50\mu A}{100\mu A/V^2 \times (0.2)^2} = \underline{\underline{25}}$$

$$\frac{W}{L} = \frac{2I_{D,tail}}{\mu_n C_{ox} (V_{ov})^2} = \frac{2 \times 100\mu A}{230\mu A \times (0.2)^2} = \underline{\underline{21.739}}$$

$$\Rightarrow \frac{W}{L} / M_0 = 21.739$$

$$\Rightarrow M_5 = 250$$

$$M_1 = M_2 = 10.869$$

$$M_3 = 108.695$$

$$M_3 = M_4 = 25$$

For maintaining phase margin,

$$\frac{g_{m5}}{C_L} \gg \frac{2g_{m1}}{C_L} \quad \omega_{p2} \quad \omega_{GBW} \text{ (dominant pole)}$$

$$g_{m5} \gg \frac{g_{m1} C_L}{C_L} = 2g_{m1}$$

Taking $g_{m5} = 10g_{m1}$

$$\Rightarrow I_5 = \frac{g_{m5} \cdot V_{ov}}{2} = 10I_D = 500\mu A$$

$$\therefore \frac{W}{L} = \frac{2 \times 500\mu A}{230\mu A/V^2 \times (0.2)^2} = \underline{\underline{108.695}}$$

$$\therefore \frac{W}{L} = \frac{2 \times 500\mu A}{100\mu A/V^2 \times (0.2)^2} = \underline{\underline{250}}$$

Taking $L = 0.5 \mu\text{m}$ for all MOSFETs.

Hand Calculated

Tail source: $M_0 = \frac{1}{2} \times 21.739 \mu\text{m} = 10.869 \mu\text{m}$

Input Pair: $M_1 = M_2 = \frac{1}{2} \times 10.869 \mu\text{m} = 5.434 \mu\text{m}$

Active Load: $M_3 = M_4 = \frac{1}{2} \times 25 \mu\text{m} = 12.5 \mu\text{m}$

Stage 2 Amp: $M_5 = \frac{1}{2} \times 250 \mu\text{m} = 125 \mu\text{m}$

Stage 2 Source: $M_6 = \frac{1}{2} \times 108.695 \mu\text{m} = 54.347 \mu\text{m}$

Current Source: $M_7 = \frac{1}{2} \times 2.173 \mu\text{m} = 1.086 \mu\text{m}$

$L = 0.5 \mu\text{m}$ for all transistors

Simulated Design Tabulation

a) Transistor Operating Points

Name & Role	Type	W (μm)	L (μm)	Target I (μA)	Simulated $V_{DS} - V_{TH}$ (V)	Simulated g_m ($\mu\text{A/V}$)	Simulated r_o (k Ω)	Simulated I (μA)
M_0 (Input stage tail)	NMOS	10.2	0.5	100	0.2	967	1.034	100
M_1 (Differential Pair) = M_2	NMOS	5.6	0.5	50	0.2	521	1.919	50
M_3 (Active Load) = M_4	PMOS	12.7	0.5	50	0.2	382	2.617	50
M_5 (Stage-2 Amplifier)	PMOS	127.2	0.5	500	0.2	383 $\times 10$	0.261	500
M_6 (Stage-2 Source)	NMOS	57.3	0.5	500	0.2	523 $\times 10$	0.191	500
M_7 (Current Source Transistor)	NMOS	1.1	0.5	10	0.2	104	9.615	10

Output Voltage swinging from 0.9V to 1.3V $\Rightarrow$ 0.4V
Input Voltage swinging from 0.9V to 1.1V $\Rightarrow$ 0.2V

$$\text{Gain} = \frac{0.4}{0.2} = \underline{2}$$

$\Rightarrow$ Hence successfully achieved gain of 2.

Values of other components

Bias reference transistor: NMOS, $W = 1.086 \mu\text{m}$, $L = 0.5 \mu\text{m}$

DC gain of the op-amp

Simulated Value: 67 dB

Verification: Using AC analysis sweep from 1 Hz ~~from~~ to 1 GHz

Power Consumption

$$\begin{aligned}\text{Total current drawn} &= I_{\text{ref}} + I_{\text{tail}} + I_{\text{stage-2}} \\ &= 10 \mu\text{A} + 100 \mu\text{A} + 500 \mu\text{A} \\ &= 610 \mu\text{A}\end{aligned}$$

$$\text{Total Power} = 1.8 \text{ V} \times 610 \mu\text{A} = 1098 \mu\text{A}$$

Simulation Plots

Fig 1: Open-Loop AC Gain Response

Observation: The magnitude plot confirms DC gain is constant at approximately 67dB for low frequencies

At unity gain frequency, there is a phase margin of 65° demonstrating high stability.

Fig 2: Closed Loop Transient Response

Observation: The OTA was configured as a non-inverting amplifier with a closed loop gain of 2 using matched $10k\Omega$ feedback resistors.

0.2V input step (0.9-1.1V) with 0.1ns rise/fall times gives
a 0.4V output step (0.9-1.3V).